

```

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-->

```

NSCopying

A protocol that objects adopt to provide functional copies of themselves.

...

protocol NSCopying

Overview

The exact meaning of “copy” can vary from class to class, but a copy must be a functionally independent object with values identical to the original at the time the copy was made. A copy produced with `[`NSCopying`](/documentation/Foundation/NSCopying)` is implicitly retained by the sender, who is responsible for releasing it.

`[`NSCopying`](/documentation/Foundation/NSCopying)` declares one method, `[`copy(with:)`](/documentation/Foundation/NSCopying/copy(with:))`, but copying is commonly invoked with the convenience method `<doc://com.apple.documentation/documentation/ObjectiveC/NSObject-swift.class/copy()>`. The `<doc://com.apple.documentation/documentation/ObjectiveC/NSObject-swift.class/copy()>` method is defined for all objects inheriting from `<doc://com.apple.documentation/documentation/ObjectiveC/NSObject-swift.class>` and simply invokes `[`copy(with:)`](/documentation/Foundation/NSCopying/copy(with:))` with the default zone.

Your options for implementing this protocol are as follows:

- Implement `[`NSCopying`](/documentation/Foundation/NSCopying)` using `<doc://com.apple.documentation/documentation/ObjectiveC/NSObject-swift.class/alloc>` and ``init...`` in classes that don't inherit `[`copy(with:)`](/documentation/Foundation/NSCopying/copy(with:))`.
- Implement `[`NSCopying`](/documentation/Foundation/NSCopying)` by invoking the superclass's `[`copy(with:)`](/documentation/Foundation/NSCopying/copy(with:))` when ``NSCopying`` behavior is inherited. If the superclass implementation might use the `[`NSCopyObject`](/documentation/Foundation/NSCopyObject)` function, make expl

icit assignments to pointer instance variables for retained objects.
- Implement `[`NSCopying`](/documentation/Foundation/NSCopying)` by retaining the original instead of creating a new copy when the class and its contents are immutable.

If a subclass inherits `[`NSCopying`](/documentation/Foundation/NSCopying)` from its superclass and declares additional instance variables, the subclass has to override `[`copy(with:)`](/documentation/Foundation/NSCopying/copy(with:))` to properly handle its own instance variables, invoking the superclass's implementation first.

Topics

Copying

`[`- copyWithZone:`](/documentation/Foundation/NSCopying/copy(with:))`

Returns a new instance that's a copy of the receiver.

Relationships

Conforming Types

`[`NSNumber`](/documentation/Foundation/NSNumber)`

`[`NSMutableDictionary`](/documentation/Foundation/MutableDictionary)`

`[`UnitArea`](/documentation/Foundation/UnitArea)`

`[`NSURLSessionDataTask`](/documentation/Foundation/NSURLSessionDataTask)`

`[`UnitElectricResistance`](/documentation/Foundation/UnitElectricResistance)`

`[`NSRegularExpression`](/documentation/Foundation/RegularExpression)`

`[`NSURLSessionWebSocketTask`](/documentation/Foundation/NSURLSessionWebSocketTask)`

`[`NSDateInterval`](/documentation/Foundation/NSDateInterval)`

`[`NSMutableOrderedSet`](/documentation/Foundation/MutableOrderedSet)`

`[`NSDate`](/documentation/Foundation/NSDate)`

`[`NSURLSessionStreamTask`](/documentation/Foundation/NSURLSessionStreamTask)`

`[`NSException`](/documentation/Foundation/Exception)`

`[`UnitFrequency`](/documentation/Foundation/UnitFrequency)`

`[`PersonNameComponentsFormatter`](/documentation/Foundation/PersonNameComponentsFormatter)`

`[`NSItemProvider`](/documentation/Foundation/ItemProvider)`

`[`Formatter`](/documentation/Foundation/Formatter)`

`[`UnitFuelEfficiency`](/documentation/Foundation/UnitFuelEfficiency)`

`[`RelativeDateTimeFormatter`](/documentation/Foundation/RelativeDateTimeFormatter)`

`[`XMLElement`](/documentation/Foundation/XML/Element)`

[`NSExtensionItem`](/documentation/Foundation/NSExtensionItem)

[`UnitEnergy`](/documentation/Foundation/UnitEnergy)

[`XMLDTDNode`](/documentation/Foundation/XMLDTDNode)

[`NSPointerFunctions`](/documentation/Foundation/NSPointerFunctions)

[`URLSessionUploadTask`](/documentation/Foundation/URLSessionUploadTask)

[`UnitPressure`](/documentation/Foundation/UnitPressure)

[`NSMutableCharacterSet`](/documentation/Foundation/NSMutableCharacterSet)

[`NSFileSecurity`](/documentation/Foundation/NSFileSecurity)

[`UnitElectricCurrent`](/documentation/Foundation/UnitElectricCurrent)

[`UnitLength`](/documentation/Foundation/UnitLength)

[`MassFormatter`](/documentation/Foundation/MassFormatter)

[`NSDictionary`](/documentation/Foundation/NSDictionary)

[`UnitTemperature`](/documentation/Foundation/UnitTemperature)

[`EnergyFormatter`](/documentation/Foundation/EnergyFormatter)

[`NSOrthography`](/documentation/Foundation/NSOrthography)

[`DateFormatter`](/documentation/Foundation/DateFormatter)

[`NSDataDetector`](/documentation/Foundation/NSDataDetector)

[`NSCalendar`](/documentation/Foundation/NSCalendar)

[`NSMeasurement`](/documentation/Foundation/NSMeasurement)

[`NSPointerArray`](/documentation/Foundation/NSPointerArray)

[`NSMutableArray`](/documentation/Foundation/NSMutableArray)

[`Port`](/documentation/Foundation/Port)

[`NSMapTable`](/documentation/Foundation/NSMapTable)

[`NSSet`](/documentation/Foundation/NSSet)

[`NSHashTable`](/documentation/Foundation/NSHashTable)

[`NSCharacterSet`](/documentation/Foundation/NSCharacterSet)

[`NSTextCheckingResult`](/documentation/Foundation/NSTextCheckingResult)

[`URLCredential`](/documentation/Foundation/URLCredential)

[`UnitAcceleration`](/documentation/Foundation/UnitAcceleration)

[`NSURLRequest`](/documentation/Foundation/NSURLRequest)

[`NSComparisonPredicate`](/documentation/Foundation/NSComparisonPredicate)

[`NSArray`](/documentation/Foundation/NSArray)

[`LengthFormatter`](/documentation/Foundation/LengthFormatter)

[`URLSessionTask`](/documentation/Foundation/URLSessionTask)

[`NumberFormatter`](/documentation/Foundation/NumberFormatter)

[`SocketPort`](/documentation/Foundation/SocketPort)

[`NSOrderedSet`](/documentation/Foundation/NSOrderedSet)

[`UnitInformationStorage`](/documentation/Foundation/UnitInformationStorage)

[`CachedURLResponse`](/documentation/Foundation/CachedURLResponse)

[`DateComponentsFormatter`](/documentation/Foundation/DateComponentsFormatter)

[`UnitElectricPotentialDifference`](/documentation/Foundation/UnitElectricPotentialDifference)

[`NSNumber`](/documentation/Foundation/NSNumber)

[`EnergyKit`](/documentation/Foundation/UnitEnergy/EnergyKit)

[`NSLocale`](/documentation/Foundation/NSLocale)

[`XMLDocument`](/documentation/Foundation/XMLDocument)

[`UnitConcentrationMass`](/documentation/Foundation/UnitConcentrationMass)

[`UnitDuration`](/documentation/Foundation/UnitDuration)

[`NSMutableSet`](/documentation/Foundation/NSMutableSet)

[`ISO8601DateFormatter`](/documentation/Foundation/ISO8601DateFormatter)

[`NSMutableString`](/documentation/Foundation/NSMutableString)

[`NSPredicate`](/documentation/Foundation/NSPredicate)

[`UnitDispersion`](/documentation/Foundation/UnitDispersion)

[`NSData`](/documentation/Foundation/NSData)

[`NSMutableData`](/documentation/Foundation/NSMutableData)

[`NSIndexPath`](/documentation/Foundation/NSIndexPath)

[`NSURL`](/documentation/Foundation/NSURL)

[`Dimension`](/documentation/Foundation/Dimension)

[`NSExpression`](/documentation/Foundation/NSExpression)

[`NSUserNotificationAction`](/documentation/Foundation/NSUserNotificationAction)

[`NSPurgeableData`](/documentation/Foundation/NSPurgeableData)

[`DateIntervalFormatter`](/documentation/Foundation/DateIntervalFormatter)

[`UnitMass`](/documentation/Foundation/UnitMass)

[`URLProtectionSpace`](/documentation/Foundation/URLProtectionSpace)

[`NSAppleScript`](/documentation/Foundation/NSAppleScript)

[`NSMachPort`](/documentation/Foundation/NSMachPort)

[`NSUUID`](/documentation/Foundation/NSUUID)

[`NSAffineTransform`](/documentation/Foundation/NSAffineTransform)

[`MessagePort`](/documentation/Foundation/MessagePort)

[`NSIndexSet`](/documentation/Foundation/NSIndexSet)

[`UnitVolume`](/documentation/Foundation/UnitVolume)

[`NSString`](/documentation/Foundation/NSString)

[`NSError`](/documentation/Foundation/NSError)

[`XMLNode`](/documentation/Foundation/XMLNode)

[`NSMutableAttributedString`](/documentation/Foundation/NSMutableAttributedString)

[`NSAppleEventDescriptor`](/documentation/Foundation/NSAppleEventDescriptor)

[`NSAttributedString`](/documentation/Foundation/NSAttributedString)

[`MeasurementFormatter`](/documentation/Foundation/MeasurementFormatter)

[`Unit`](/documentation/Foundation/Unit)

[`NSMutableIndexSet`](/documentation/Foundation/NSMutableIndexSet)

[`NSNull`](/documentation/Foundation/NSNull)

[`URLSessionDownloadTask`](/documentation/Foundation/URLSessionDownloadTask)

[`NSDecimalNumber`](/documentation/Foundation/NSDecimalNumber)

[`NSNotification`](/documentation/Foundation/NSNotification)

[`ByteCountFormatter`](/documentation/Foundation/ByteCountFormatter)

[`NSSortDescriptor`](/documentation/Foundation/NSSortDescriptor)

[`NSUserNotification`](/documentation/Foundation/NSUserNotification)

[`UnitAngle`](/documentation/Foundation/UnitAngle)

[`NSCompoundPredicate`](/documentation/Foundation/NSCompoundPredicate)

[`NSCountedSet`](/documentation/Foundation/NSCountedSet)

[`HTTPURLResponse`](/documentation/Foundation/HTTPURLResponse)

[`UnitIlluminance`](/documentation/Foundation/UnitIlluminance)

[`NSMutableURLRequest`](/documentation/Foundation/NSMutableURLRequest)

[`UnitPower`](/documentation/Foundation/UnitPower)
[`UnitSpeed`](/documentation/Foundation/UnitSpeed)
[`Scanner`](/documentation/Foundation/Scanner)
[`NSPersonNameComponents`](/documentation/Foundation/NSPersonNameComponents)
[`ListFormatter`](/documentation/Foundation/ListFormatter)
[`UnitElectricCharge`](/documentation/Foundation/UnitElectricCharge)
[`NSURLQueryItem`](/documentation/Foundation/NSURLQueryItem)
[`NSDateComponents`](/documentation/Foundation/NSDateComponents)
[`URLResponse`](/documentation/Foundation/URLResponse)
[`XMLDTD`](/documentation/Foundation/XMLDTD)
[`URLSessionConfiguration`](/documentation/Foundation/URLSessionConfiguration)
[`NSTimeZone`](/documentation/Foundation/NSTimeZone)
[`NSURLComponents`](/documentation/Foundation/NSURLComponents)

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